Python Assignment
Topic-1: Introduction to Python

Q1. Introduction to Python and its Features.

Ans.

Python is a popular programming language used by students and professionals. It is simple and easy to understand. Python is a high-level language, so we can write code in easy English format. It is also an interpreted language which means code runs line by line. Python is used in web development, data science, automation, and artificial intelligence. It supports object-oriented programming and has many built-in libraries. Python is also platform independent, so it works on different systems. Because of its simplicity, beginners prefer Python for learning programming.

Example:

```
print("Hello Anshu")
```

Q2. History and evolution of Python.

Ans.

Python was created by Guido van Rossum in the late 1980s. It was officially released in 1991. The main goal was to make a simple and readable programming language. Over time, Python became very popular because of its easy syntax. Different versions of Python were released with new features. Python 2 was widely used earlier, but now Python 3 is used. Python keeps improving with updates and strong community support. Today, it is one of the top programming languages in the world.

Q3. Advantages of using Python over other programming languages.

Ans.

Python has many advantages compared to other languages. It is easy to learn and use, especially for beginners. Python requires less code, so programs are shorter. It is an interpreted language, so errors can be found easily. Python is platform independent and works on different operating systems. It also provides a large number of built-in libraries. Python is used in many fields like web development, AI, and data analysis. These advantages make Python a powerful and flexible language.

Q4. Installing Python and setting up the development environment.

Ans.

To start Python, we first need to install it on our system. We can download Python from the official website. After downloading, run the setup file and install it. It is important to select 'Add Python to PATH' option. After installation, we can use tools like IDLE, VS Code, or PyCharm to write code. These tools help in writing and running Python programs easily. After installation, we can check it using the command python --version. This confirms that Python is installed successfully.

Q5. Writing and executing your first Python program.

Ans.

Writing the first Python program is very easy. We just need to open any editor like IDLE or VS Code. Then we write a simple program using the print() function. After saving the file with .py extension, we can run the program. Python executes the code line by line and shows output. This helps beginners understand how Python works. The first program is usually used to display a simple message on the screen.

Example:

```
print("This is my first Python program")
```